

# Lesson 3: The Value of Privacy

45 minutes

## Overview

Students develop their own opinions on the privacy tradeoffs inherent in many modern computing innovations. At the beginning of the lesson students watch a video on facial recognition technology that presents the tradeoffs between convenience and privacy and asks them to determine whether they think the tradeoff is worth it. Students respond to two videos that look at different tradeoffs between privacy, security, and convenience. Students then evaluate the website or app they investigated in the previous lesson to determine if they think the benefits of the service outweigh the privacy risks. At the end of the class students discuss whether they generally think convenience outweighs privacy concerns.

## Standards

Full Course Alignment

### CSP Conceptual Framework

- ▶ **IOC-1** - While computing innovations are typically designed to achieve a specific purpose, they may have unintended consequences.
- ▶ **IOC-2** - The use of computing innovations may involve risks to your personal safety and identity.

### CSTA K-12 Computer Science Standards (2017)

- ▶ **IC** - Impacts of Computing

## Agenda

Warm Up (5 minutes)

Activity (35 minutes)

Convenience vs Privacy - Facial Recognition

Security vs Privacy - Should the Government Need a Warrant?

Privacy Tradeoffs - Your Innovation

Wrap Up (5 minutes)

Assessment

## Objectives

Students will be able to:

- Assess a computing innovation to identify the specific privacy risks that could arise from the data it collects and stores.
- Evaluate whether the benefits to society from a given computing innovation outweigh the privacy risks it poses.
- Explain the risks to privacy that arise from using modern computing technology

## Preparation

- Review the videos to make sure you are prepared to lead the subsequent discussions
- Make sure you have access to the activity guides students completed in the previous lesson
- Check the **"Teacher's Lounge"** forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our **Virtual Lesson Modifications**

## Links

**Heads Up!** Please make a copy of any documents you plan to share with students.

For the teachers

- **The Value of Privacy** - Slides

▼ Make a Copy

For the students

- [Cops Need A Warrant To Search Your Cell Phone's Location History.](#) - Video
- [Is Facial Recognition Invading Your Privacy?](#) - Video
- [Privacy, Security, and Innovation](#) - Activity Guide [▼ Make a Copy](#)

## Teaching Guide

### Warm Up (5 minutes)

💡 Teaching Tip ▲

**Move Quickly to the Main Activity:** An optional prompt is provided but otherwise move quickly into watching and reviewing the videos so that students will have enough time to complete their activity guides.

**Distribute:** As students walk in give them copies of their [Privacy, Security, and Innovation](#) which they began filling out in the previous lesson.

#### 🎤 Remarks

Review your activity guide from last class. Remind yourself what your privacy concerns were for the app / website you chose.

### Activity (35 minutes)

#### Convenience vs Privacy - Facial Recognition (10 minutes)

As we watch the following video on facial recognition takes notes on:

- What are the benefits of the technology? What does it make more convenient/fun?
- What are the privacy concerns that arise?
- Why would governments or businesses be interested in this technology?

Afterwards we'll discuss whether we think the benefits outweigh the privacy concerns.

**📺 Display:** Show the video about facial recognition technology.

**🗨️ Discuss:** *Students should share notes with their partner about the three prompts and then discuss the provided question "Do you believe the privacy risks posed by facial recognition technology outweigh the benefits/convenience?" Have students share some of their thoughts as a class.*

**Discussion Goal:** Use this first video to focus on the tradeoffs between convenience and privacy. Students have likely used or seen new technology that uses facial recognition that might be helpful or fun, and many examples are provided in the video. The following video focuses more on the security vs. privacy balance.

## Security vs Privacy - Should the Government Need a Warrant? (10 minutes)

As we watch the following video on location data, take notes on the following:

- What kind of data is being collected and shared?
- Who is the "third party" in this situation?
- Why is this a tradeoff between privacy and security?

 **Display:** Show the video about location data.

 **Discuss:** Students should share notes with their partner about the three prompts and then discuss the provided question *"Do you agree with the Supreme Court's ruling? Do you think it should apply to other kinds of data?"* Have students share some of their thoughts as a class.

**Discussion Goal:** Use this discussion to build off the theme's discussed after the previous video about facial recognition. Push students to think of other types of data that third-parties collect about them and how they think that data should be handled by those parties and the government.

## Privacy Tradeoffs - Your Innovation (15 minutes)

Students are given a good chunk of class time to complete this activity. Ask students to reflect quietly but as individuals or groups finish they should begin discussing with one another. If most students finish early you could begin a full-class discussion.

Individually fill in the last page of your activity guide.

### Teaching Tip

**Developing an Informed Opinion:** This lesson has asked students to hold off on weighing in on the innovations they researched in the previous class to develop a more nuanced and informed opinion about the tradeoffs between innovation, privacy, and security. As students move to this last activity, remind them of the more nuanced opinion they've developed, and ask them to reflect on how the different examples they've seen impact their overall opinions on this tradeoff.

## Wrap Up (5 minutes)

 **Discuss:** *Based on the discussions we had today, which of the following best describes your overall feelings with sharing private data in order to power innovations or ensure security.*

- 1: Totally Comfortable: "Have my data! It makes the technology I love work and I have nothing to hide!"
- 2: Mostly Comfortable: "I want tech innovations and stronger security. Let's make sure we take care of the most damaging privacy concerns."
- 3: Mixed: "There's a lot of this that makes me uncomfortable, but I'm still going to use technology."
- 4: Mostly Uncomfortable: "Privacy is more important than empowering innovations or ensuring security. I would give up on some tech innovations to ensure my privacy"
- 5: Totally Uncomfortable: "I'd give up most technology and would like to see much stronger limits on what kind of data can be collected and stored, even if it limits the introduction of new technology"

Have a few people from each end of the spectrum share their reasoning.

**Discussion Goal:** Use this discussion to do some classwide synthesis of today's discussions about the tradeoffs between privacy and empowering new technology or ensuring security. The goal here is to move individual students to think broadly about their opinions, not force every student to have the same opinion. It's not necessary

that students agree, but prompt them to think critically about this important question posed by modern technology. Some ways to do this:

- Ask them why they hold their opinions and ask them to tie them to today's lesson
- Ask them to give specific examples
- Ask students who disagree to politely debate their opinions
- Ask students to tie their opinions to their own experiences

 **Review:** Review the key takeaways at the end of the lesson.

## *Remarks*

There will always be tradeoffs between privacy and new exciting technology. Think about when you use a social media site and are offered targeted advertising based on your browser history. This can be useful! You may be introduced to things you want to buy or information you want to know about. However, it can also be harmful. Targeted advertising can be used to gather data about a user or a group of users for malicious reasons. In this case, this innovation is not being used as it was originally intended to be used.

Consider this as you research the impacts of your computing innovations for the Innovation Simulation project.

---

## Assessment

**Activity Guide:** Once again collect the activity guide about the app / computing innovation that students researched in the previous lesson. If students missed the previous lesson they should use information collected by a partner about an app. A rubric included on the bottom of the activity guide can be used for assessment.



This work is available under a **Creative Commons License (CC BY-NC-SA 4.0)**.

If you are interested in licensing CodeAI materials for commercial purposes **contact us**.